

Boundary conditions, initial conditions, and solver:

EXHALE / EXHALE vs. ATES, Salz+2016,

Kubyshkina+2018, and Murray-Clay+2009

A focused comparison of the boundary conditions (BC), initial conditions (IC), and steady-state solver of EXHALE / EXHALE against the relevant 1-D atmospheric-escape codes: its own parent code ATES (Caldirola et al. 2021), the PLUTO–CLOUDY interface (TPCI) of Salz et al. (2016), the grid code of Kubyshkina et al. (2018) (numerical scheme from Erkaev et al. 2016), and the steady-state boundary-value archetype of Murray-Clay et al. (2009). The aim is to place EXHALE’s design choices in context — what it inherited from ATES, what it added, which choices are field-standard, and which capabilities the other codes have that EXHALE lacks. A companion design sketch (auto_ic_design.md) follows up on the main actionable finding: a Kubyshkina-style automatic IC selector. Full BC/IC details of EXHALE itself are in EXHALE_BC_and_IC.tex.

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1. The three codes at a glance

The headline structural fact is that **EXHALE and Salz are numerical near-twins** — same reconstruction, same Riemann solver, same time integrator — so their differences are in the BC/IC details and in the microphysics, not in the hydrodynamics. Kubyshkina is the outlier: a deliberately simpler, faster scheme whose strengths lie in robustness

	Salz+2016 (TPCI)	Kubyshkina+2018	EXHALE
Hydro scheme	WENO3 + HLLC + RK3 (Godunov)	McCormack predictor–corrector (finite diff., no Riemann solver)	PLM/WENO3 + HLLC + RK (Godunov)
Source term	operator-split forward Euler	in-scheme	semi-implicit energy + Brent T -solve
Steady state	time relaxation only	time relaxation	time relaxation + JFNK Newton finish
Microphysics	CLOUDY full non-equilibrium photoionization (separate auto grid, 600–1200 cells)	H+H ₂ network, Ly α +H ₃ ⁺ cooling, $\eta = 15\%$ fixed	ionization equilibrium (H/He+metals, MINPACK), CHIANTI metal cooling, He 2 ³ S, H($n=2$)/Ly α RT
Composition	H/He (metals/molecules in CLOUDY only)	pure H; H ₂ molecular at base	H/He + 10 metals (self-consistent EOS)
Grid	stretched 500 cells, 1–12/15 R_{pl} , base $\Delta r=2\times 10^{-4} R_{\text{pl}}$	1 $R_{\text{pl}} \rightarrow$ Roche radius	500 cells + 2 ghost, 1 $R_{\text{pl}} \rightarrow r_{\text{max}}$
Potential	planet+star+centrifugal (2-body) + radiation pressure + conduction	Roche (Erkaev 2007) + conduction	Roche / spherical

Table 1: High-level comparison. EXHALE and Salz share the same Godunov-type numerics (WENO3+HLLC+RK); Kubyshkina uses a lower-order, more diffusive finite-difference scheme optimized for mass-producing a ~ 7000 -model grid.

across escape regimes and in grid-production throughput. Murray-Clay is a different beast entirely — a steady-state boundary-value solver rather than a time-relaxation code (§5.3).

2. Heritage: what EXHALE inherited from ATES (Caldirola+2021)

EXHALE / EXHALE is a fork of ATES, so its BC/IC/solver baseline is literally Caldirola et al. (2021). Reading that paper makes the inheritance explicit and isolates what is genuinely new in EXHALE.

Inherited verbatim (or nearly so):

- *Lower BC.* Density and temperature fixed at R_{pl} , total $n = 10^{14} \text{ cm}^{-3}$ (the same value Salz uses), T at the planet’s equilibrium temperature. *The base velocity valve:* “the velocity in the ghost cells at the lower boundary was obtained through a zeroth-order extrapolation from the first computational cell in the case of inflow characteristics, while it was set to zero otherwise” — i.e. EXHALE’s $\max(v_1, 0)$ valve is the original ATES valve. Caldirola even states “ATES employs the same boundary conditions as TPCI at the planet radius,” so the ATES–Salz BC equivalence noted in §3. is asserted in the ATES paper itself.
- *Upper BC.* Zero-gradient (PLM) / linear extrapolation (ESWENO3) at the Roche-lobe radius — exactly EX-

HALE’s outer BC.

- *IC.* “A fully neutral, isothermal sphere of H and He ... with a linear initial velocity profile $v(t=0) = (r-1)/2$ ” — this is EXHALE’s cold-hydrostatic default IC.
- *Solver core.* Godunov finite-volume, SSPRK3 time integrator, HLLC default (+Roe, LLF), PLM and ESWENO3 reconstruction, $C_{\text{CFL}} = 0.6$, 500-cell mixed uniform–stretched grid, and the *two-stage PLM $\rightarrow$ ESWENO3 relaxation* with the final tolerance $\Delta\dot{M}/\dot{M} < 10^{-3}$. Ionization solved in photoionization equilibrium via the MINPACK dogleg method; ion advection only as a post-process. All of this is EXHALE’s backbone.

Added by EXHALE (none of these are in Caldiroli+2021): the trace-metal network and its self-consistent EOS contribution (eos_metals); He 2^3S , H($n=2$) and Ly α RT; CHIANTI metal-line cooling; the *JFKN Newton steady-state finish* (ATES has *no* implicit steady-state solver — it is time-relaxation only); the semi-implicit energy + Brent T -solve; the *smooth valve* (Valve eps) and the *transonic* and *hot-Parker* IC families; the dynamic base-pressure correction δp_{bc} . In short, EXHALE keeps ATES’s hydro core and BC/IC skeleton and adds the metal physics, the excited-H/Ly α physics, and a genuine steady-state solver on top. Notably, Caldiroli+2021 makes *no mention of base oscillations or “breathing”*; that phenomenology and its mitigations (smooth valve, PP-adv guard) are EXHALE-era findings.

3. Boundary conditions

3.1 Lower boundary

Salz. Two quantities fixed at the bottom — *density and pressure* — and *none* at the top, which the paper notes is “the correct choice for a subsonic inflow and a supersonic outflow.” The base is anchored at $n \approx 10^{14} \text{ cm}^{-3}$, $p \approx 14 \mu\text{bar}$ (set from T_{eq}). Crucially, *the boundary-cell velocity is copied from the first interior cell each step but only inflow (wind-feeding) is allowed; otherwise it is set to zero* — a choice the paper explicitly introduces “to dampen oscillations that appear with steep density gradients at the lower boundary.”

This is **exactly EXHALE’s one-way velocity valve** $v_{\text{gh}} = \max(v_1, 0)$, adopted independently for the same reason (the breathing base, EXHALE_BC_and_IC.tex §2.4). EXHALE layers three refinements on top that Salz does not have: (i) the dynamic pressure correction δp_{bc} that pins the base to $T = T_0$ exactly through the EOS; (ii) the smooth valve (Valve eps) that makes the kink differentiable for the Newton solver; (iii) the metal mass/electron terms in the base density and pressure (eos_metals, 2026-06-13).

Kubyshkina (scheme from Erkaev+2016). Lower boundary at the photospheric radius R_{pl} with *temperature fixed to T_{eq} and pressure set from Fossati+2017*, composition *molecular H_2* . The underlying Erkaev+2016 model is explicit: *five* Dirichlet conditions at the base ($\tilde{T}=1$, $\tilde{p}=1$, and the hydrogen speciation fixed to pure H_2), with the base placed at the *homopause* (base of the thermosphere, $n_{\text{H}_2} = 5 \times 10^{12} \text{ cm}^{-3}$), not the photosphere; the velocity is *not* fixed at the base. (Note the potential differs between the two papers: Erkaev+2016 uses a *point-mass* planetary potential, while Kubyshkina+2018 adds the Roche term.)

Versus EXHALE: EXHALE fixes (ρ, p) with p dynamically corrected so that $T = T_0$, i.e. *effectively the same pair (ρ, T)* . The substantive difference is the **H_2 molecular base**: it places the lower boundary deeper, in a near-hydrostatic

molecular layer, which is what lets Kubyshkina follow boil-off (low-gravity, nearly hydrostatic but rapidly escaping) atmospheres smoothly. EXHALE’s atomic H/He base does not represent that layer and is correspondingly weaker there — the same regime where EXHALE’s breathing base is hardest to converge.

3.2 The base *radius* is the real lever, not the base values (Murray-Clay+2009)

Murray-Clay et al. (2009) is the standard reference for *how much the lower BC matters*, and their answer is sharp: within physically realistic ranges, the mass-loss rate is **insensitive** to the base density, temperature, and ionization fraction (so long as $\tau_{\text{base}} \gg 1$, $T_{\text{base}} \ll 10^4$ K, $f_+ \ll 1$) — they call the solution “quasi-unique.” What $\dot{M}$ is strongly sensitive to is the base *radius* r_{min} : $\dot{M} \propto R_{\text{pl}}^3$, so a 20% change in r_{min} moves $\dot{M}$ by up to a factor of two. Salz independently finds the same insensitivity to the boundary *temperature* ($< 5\%$ change in $\dot{M}$ for a 40% temperature change).

This reframes the whole lower-BC discussion and applies directly to EXHALE: the precise values of ρ_{bc} , T_0 , and the new metal EOS terms are *not* the lever (consistent with the eos_metals mass effect being $\sim 1\%$ in $\dot{M}$), whereas the choice of base *radius* R_0 is. This echoes EXHALE’s own WASP-121b experience, where moving the base from the 4-mbar transit radius ($R_0 = 1.766$) to the correct 1- μ bar level ($R_0 = 2.058$) was what brought Case A into agreement with Huang+2023 — the same “base-radius is the lever” effect Murray-Clay quantify. Practical takeaway: spend effort getting R_0 (and the pressure level it corresponds to) right; do not over-tune ρ_{bc} or T_0 .

3.3 Upper boundary

All three are essentially identical: **supersonic free outflow**, imposed as zero-gradient / linear extrapolation with no fixed condition (Kubyshkina: “radial derivatives become zero” at the Roche radius; Salz: “none at the top”; EXHALE: copy / WENO3 linear extrapolation). The same well-posedness caveat applies to all three when the flow is not actually supersonic at the outer radius (deep Roche-lobe overflow); EXHALE documents this explicitly (EXHALE_BC_and_IC.tex §3).

4. Initial conditions

	Salz	Kubyshkina	EXHALE
Method	single fixed IC (tuned to a medium- $\dot{M}$ atmosphere), same for all planets	automatic selection of an initial profile for each planet	cold hydrostatic (default) / transonic / hot-Parker
Stated property	“final steady state does not depend on the IC”	auto-selection accelerates a 7000-model grid	same $\dot{M}$ regardless; convergence is momentum- (du) -gated
Transient	two shock waves cross in the first hours, then settle	—	breathing-base limit cycle

Table 2: Initial-condition strategies.

Salz demonstrates IC-independence of the steady state and gets away with a single hand-tuned IC, at the cost of an

expensive shock-laden relaxation. **Kubyshkina’s automatic IC selection for each planet is the key engineering idea:** choosing an appropriate starting atmosphere at each grid point is what makes mass-producing 7000 models tractable and what lets the code launch winds across the full escape-parameter range, including the boil-off cases that do not launch from a cold static start. EXHALE already has the *ingredients* for this — the transonic and hot-Parker IC families exist precisely for “won’t launch from hydrostatic” cases — but the choice is currently *manual* (input keys), whereas Kubyshkina automates it. This is the main actionable gap; see `auto_ic_design.md`.

EXHALE’s IC behavior is, however, the best-documented of the three: the benchmark (`initial_condition_benchmark.pdf`) quantifies that a naive full-Parker density seed diverges (self-amplifying breathing) while the warm-seed converges to the same $\dot{M}$ with no speedup — because convergence is gated by the slow momentum relaxation, not the thermal/ionization relaxation a warm seed shortcuts.

5. Solver: differences, strengths, weaknesses

5.1 Versus Salz (the close comparison)

Shared: WENO3/HLLC/RK family, the one-way valve, IC-independent steady state.

EXHALE advantages:

- *JFNK Newton steady-state finish.* Salz relaxes in time only, so its convergence criterion is $\Delta\dot{M}/\dot{M} < 0.1$ (10%) with residual oscillations handled by time-averaging. EXHALE converges to the fixed point directly ($\|R\|_\infty \rightarrow 0$), a much stricter standard.
- *Metals self-consistent in the hydro EOS* (2026-06-13). Salz’s published runs keep metals/molecules inside CLOUDY’s separate grid and run H/He-only hydro.
- *Semi-implicit energy + Brent T-solve* avoids the hot-root trap of stiff metal cooling.

Salz advantages:

- *CLOUDY’s full non-equilibrium photoionization with self-consistent radiative transfer and molecular/metal cooling* — fundamentally more accurate microphysics than EXHALE’s “ionization equilibrium + prescribed cooling fits,” especially in the radiative-equilibrium base layers.
- Very high base resolution ($\Delta r \sim 2 \times 10^{-4} R_{\text{pl}}$) suppresses the pressure-gradient cancellation error that drives low- $\dot{M}$ oscillations.
- Cost: TPCI is expensive and has no steady-state solver.

5.2 Versus Kubyshkina

EXHALE advantages:

- *HLLC Riemann + WENO3* vs. the McCormack predictor–corrector $\Rightarrow$ much lower numerical diffusion. (Salz attributes the grid-dependent $\dot{M}$ of older Lax–Friedrich-type schemes, e.g. Tian et al. 2005, to exactly this diffusion; EXHALE/Salz are free of it, Kubyshkina’s scheme is comparatively diffusive.)

- Far richer microphysics (metals, He 2^3S , Balmer/Ly α RT) vs. $\text{H}/\text{H}_2 + \text{Ly}\alpha + \text{H}_3^+$ with a fixed $\eta = 15\%$.

Kubyshkina advantages:

- *Robust across the full boil-off / blow-off range.* The molecular- H_2 base plus automatic IC for each planet let it span escape parameters $\Lambda \sim 20\text{--}80$ and produce 7000 models — precisely the low-gravity / nearly-hydrostatic regime where EXHALE’s breathing base is hardest to converge (e.g. HD 189733 b).
- Simple and fast: ideal for the large model grids used in planetary evolution calculations. EXHALE is more accurate per case but heavier and breathing-limited.

The underlying scheme (Erkaev+2016) is a two-step MacCormack predictor–corrector, *second-order* and explicitly noted as “suitable for smooth solutions without sharp discontinuities” — i.e. it lacks the shock-capturing of EXHALE’s HLLC. It reaches steady state by time relaxation (no artificial viscosity stated), and the paper validates it against the analytic Parker (1958) wind and against Murray-Clay+2009 (3.33 vs. $3.3 \times 10^{10} \text{ g s}^{-1}$). So the EXHALE-vs-Kubyshkina solver gap is concrete: a Godunov/HLLC/WENO3 scheme versus a smooth-flow second-order finite-difference scheme.

5.3 A different solver philosophy: steady-state BVP (Murray-Clay+2009)

Murray-Clay+2009 is worth a dedicated note because it is the principal *methodological alternative* to everything above: instead of relaxing a time-dependent code to a stationary state, it solves the *steady-state* equations directly as a two-point boundary-value problem. The two points are the base and the *sonic point*; six BCs are imposed (two Parker critical-point conditions at the sonic point, four at the base). The interior ($R_{\text{pl}} \rightarrow r_s$) is solved by a Newton relaxation (solvrde) on the finite-differenced ODEs, and the exterior ($r_s \rightarrow R_{\text{Roche}}$) by Bulirsch–Stoer integration, iterated for consistency.

Two lessons for EXHALE:

- *The sonic point is the natural place to anchor a steady solver.* Murray-Clay stress that “for every transonic wind there are an infinite number of breeze solutions,” and codes that do not enforce the critical-point conditions “found, not surprisingly, breezes instead.” EXHALE’s JFNK Newton finish is the time-domain analogue of their relaxation, but it does not explicitly impose the sonic-point conditions — a possible robustness upgrade for the hard-to-launch cases, and an alternative to the transonic IC.
- *Continuation/homotopy is how they launch hard cases.* They build the final solution by “solving successively more complicated problems”: start from an isothermal wind, then add photoionization, heating, cooling, and tidal gravity “one by one ... in a diluted form first and gradually strengthened to full amplitude.” This is a proven recipe for reaching a transonic solution that EXHALE does not currently use; it is noted as an alternative IC/continuation strategy in `auto_ic_design.md`.

The trade-off: a steady-state BVP solver like Murray-Clay’s is elegant and fast for a single transonic wind, but it bakes in the transonic topology (it cannot represent a genuinely time-dependent or non-launching atmosphere) and does not naturally carry a large reaction network. EXHALE’s time-relaxation-plus-Newton approach is more general at the cost of the breathing-base difficulty.

5.4 Wind-AE (Broome et al. 2025): MC09’s successor, verified in-tree

Added 2026-06-13. The Murray-Clay code did not stay an archetype: its direct successor, **Wind-AE** (Broome et al. 2025, ApJ 995, 198; BSD-3, PyPI `wind_ae`), is available in this workspace under `wind-ae-main/`. It keeps the MC09 BVP relaxation core (Numerical Recipes `solvr` from the base to the sonic point, Bulirsch–Stoer outward) and adds atomic metals, multifrequency XUV with X-ray secondary/K-shell ionization, a toggleable tidal + centrifugal potential, and — importantly — a Python wrapper that implements exactly the continuation/homotopy machinery of §5.3 (`ramp_to`, `ramp_grav`, `ramp_Ftot`, `polish_bcs`). Limitations relative to EXHALE: one ionization stage per species (no Mg II/Fe II line work), no Fe/Ca cooling, substellar-point streamline geometry, and integration stops near the Coriolis turning radius. A Fortran port is planned in `md_backup/wind_ae_fortran_port_plan.md`; it is not *required* for use — the C core compiles and runs on this Linux machine as-is (~ 14 s per solve) and the wrapper works under Python 3.11 — but remains under consideration for code completeness/uniformity of the in-house tool set.

The following was verified hands-on (2026-06-13), ramping from the bundled MC09 fiducial to EXHALE’s own `input.inp` parameters with matched integrated XUV flux (caveat: Wind-AE’s bundled multifrequency spectrum vs. EXHALE’s power law; substellar vs. Rate/2 heating geometry — factor-of-a-few agreement is the realistic target):

- **HD 209458b** (EXHALE values $0.720 M_J$, $1.401 R_J$, $F_{\text{XUV}} = 1081 \text{ erg cm}^{-2} \text{ s}^{-1}$): converges in seconds; $T_{\text{max}} = 8640 \text{ K}$, substellar $4\pi r^2 \rho v = 10^{10.78} \text{ g s}^{-1}$ at the sonic point ($r_s = 3.49 R_{\text{pl}}$). EXHALE’s own metals-off run gives a flat $4\pi r^2 \rho v = 10^{10.84}$ and $T_{\text{max}} = 8300 \text{ K}$ — 15% / 4% agreement from a completely independent method.
- **HD 189733b** (the planet where EXHALE’s time marching limit-cycles at $du \sim 2$ and never converges): the BVP relaxation *converges cleanly* ($T_{\text{max}} = 11300 \text{ K}$, $r_s = 3.83 R_{\text{pl}}$, global $\dot{M} = 10^{10.2}$ with their 0.33 substellar-to-global factor). **A steady transonic solution exists for this planet.** This is the missing experiment from the 2026-06-08 convergence study: the HD 189733b oscillation is a property of EXHALE’s time-marching + base-BC numerics (acoustic reflection), not of the physics. The NSCBC / sonic-anchored-Newton routes are therefore worth their cost for this regime.
- **WASP-121b** (near-RLOF, $F_{\text{XUV}} \sim 1.6 \times 10^6$): the `ramp_to` continuation stalls around half the target flux (step size driven to $\sim 1\%$); the deep-RLOF + extreme-XUV corner appears not (easily) reachable — consistent with the sonic-anchored formulation’s difficulty when the critical point approaches L1. EXHALE’s Roche machinery (Phase 4) has no replacement here.

Wind-AE as an EXHALE initial condition (tested). A ~ 150 -line converter (`src/Utils/windae_to_exhale_ic.py`) interpolates a `windsoln.csv` onto the EXHALE grid (from `IC_dump.txt`), blends hydrostatically down to the EXHALE base anchor below Wind-AE’s $R_{\text{min}} \simeq 1.06 R_{\text{pl}}$, extends flux-conservingly above its R_{max} , and writes EOS-consistent `Hydro_ioniz_IC.txt` / `Ion_species_IC.txt` for Load IC. Test on HD 209458b (H/He, spherical, $10 R_{\text{pl}}$, two-stage PLM $\rightarrow$ WENO3, $du_{\text{th}} = 10^{-3}$):

- *Hazard first:* with plain marching the run “converges” in **3 steps** — a false stop. EXHALE’s du criterion measures the spatial spread of $\rho v r^2$, which a BVP solution satisfies *by construction* ($du = 2.5 \times 10^{-4}$ at load); the state is still Wind-AE’s solution, not EXHALE’s fixed point. Any Wind-AE-seeded run must use Solver: Newton

(or an R -based gate; the same trap as the $\sim 5\%$ du -stop spread documented in `auto_ic_design.md` §6.1, in its most extreme form).

- *Genuine value second*: under Newton-mode marching (no du stop) the loaded state evolves smoothly toward EXHALE's own attractor — flux level $10^{10.78} \rightarrow 10^{10.68}$, spread $2.5 \times 10^{-4} \rightarrow 6.6 \times 10^{-2}$ (peak) $\rightarrow 2.1 \times 10^{-2}$ at 146k steps, monotone, no NaN. The **cold-start control crashed** (NaN at the breathing base, $r \simeq 1.18$, step 172 469, still at $du = 3.7$) in the same configuration. For this hard, weakly-driven case the Wind-AE warm start survives where the cold start dies.

In short: Wind-AE is the working realization of both §5.3 lessons (sonic-point anchoring *and* continuation), usable today as an independent $\dot{M}$ oracle and as a warm-start IC generator for EXHALE's hardest convergence regime — with the explicit caveat that its solution must never be accepted through the bare du stop.

6. What EXHALE should take away

1. **The valve is standard, and EXHALE's version is the most complete.** Salz independently adopted the same one-way valve for the same anti-oscillation reason; EXHALE's δp_{bc} , smooth valve, and metal EOS sit on top of it. The lower BC is not a weak point.
2. **The Newton finish is a genuine, unique strength** — only EXHALE converges to the fixed point rather than time-averaging. The real limitation is not the solver but the breathing base in low- $\dot{M}$ / low-gravity cases.
3. **The most worthwhile idea to adopt is Kubyskhina's automatic IC selection** (and, relatedly, a deeper / molecular base for boil-off). EXHALE already has the transonic and hot-Parker IC families and computes the escape parameter $b_0 \equiv \Lambda = GM_{pl} m_H / (k_B T_0 R_{pl})$; what is missing is an automatic rule that picks the IC family from b_0 and the sonic-point location. This directly targets EXHALE's weakest regime. See `auto_ic_design.md`.
4. **The microphysics ceiling is Salz/CLOUDY.** The known limitations of EXHALE's equilibrium + fitted cooling (the `use_2lev_cool` branch, fine-structure floors, etc. in `TO_BE_DONE.md`) are exactly what a non-equilibrium RT treatment like TPC1 would resolve — a long-term direction, not a near-term change.
5. **Tune the base radius, not the base values** (Murray-Clay+2009; §3.2). $\dot{M}$ is quasi-insensitive to ρ_{bc} , T_0 , and the new metal-EOS terms but $\propto R_{pl}^3$ in the base radius. This is reassuring for the $\sim 1\%$ `eos_metals` change and is the lesson behind EXHALE's own $R_0 = 1.766 \rightarrow 2.058$ correction for WASP-121b. Effort on R_0 /base-pressure calibration pays off; effort on ρ_{bc}/T_0 does not.
6. **Consider anchoring the steady solver at the sonic point** (Murray-Clay+2009; §5.3). EXHALE's Newton finish does not enforce the Parker critical-point conditions; doing so (or using continuation/homotopy to reach the transonic branch) is a candidate robustness upgrade for the non-launching cases, complementary to the automatic IC selector.
7. **Wind-AE makes items 2 and 6 actionable today** (§5.4). The MC09 successor runs in-tree: it proved a steady HD 189733b solution exists (the limit cycle is numerics, not physics), cross-checks $\dot{M}$ at the 15% level on

HD 209458b, and its solutions work as warm-start ICs that survive where the cold start NaN-crashes — provided the run is finished with Solver: Newton, never the bare *du* stop.